

5. The probability of being in an odd-numbered grade is $2x + x + x = 4x = 4/7$.
6. Assume that all eight possible combinations of faces are equally likely. Only two of those combinations have all three faces the same, so the probability is $1/4$.
7. The possible genotypes of the offspring are aa and Aa , since one parent will definitely contribute an a , while the other can contribute either A or a . Since the parent who is Aa contributes each possible allele with probability $1/2$ each, the probabilities of the two possible offspring are each $1/2$ as well.
8. (a) The sample space contains 12 outcomes: (Head, 1), (Tail, 1), (Head, 2), (Tail, 2), etc.
 (b) Assume that all 12 outcomes are equally likely. Three of the outcomes have Head and an odd number, so the probability is $1/4$.

1.7 Counting Methods

Commentary

If you wish to stress computer evaluation of probabilities, then there are programs for computing factorials and log-factorials. For example, in the statistical software *R*, there are functions `factorial` and `lfactorial` that compute these. If you cover Stirling's formula (Theorem 1.7.5), you can use these functions to illustrate the closeness of the approximation.

Solutions to Exercises

1. Each pair of starting day and leap year/no leap year designation determines a calendar, and each calendar correspond to exactly one such pair. Since there are seven days and two designations, there are a total of $7 \times 2 = 14$ different calendars.
2. There are 20 ways to choose the student from the first class, and no matter which is chosen, there are 18 ways to choose the student from the second class. No matter which two students are chosen from the first two classes, there are 25 ways to choose the student from the third class. The multiplication rule can be applied to conclude that the total number of ways to choose the three members is $20 \times 18 \times 25 = 9000$.
3. This is a simple matter of permutations of five distinct items, so there are $5! = 120$ ways.
4. There are six different possible shirts, and no matter what shirt is picked, there are four different slacks. So there are 24 different combinations.
5. Let the sample space consist of all four-tuples of dice rolls. There are $6^4 = 1296$ possible outcomes. The outcomes with all four rolls different consist of all of the permutations of six items taken four at a time. There are $P_{6,4} = 360$ of these outcomes. So the probability we want is $360/1296 = 5/18$.
6. With six rolls, there are $6^6 = 46656$ possible outcomes. The outcomes with all different rolls are the permutations of six distinct items. There are $6! = 720$ outcomes in the event of interest, so the probability is $720/46656 = 0.01543$.
7. There are 20^{12} possible outcomes in the sample space. If the 12 balls are to be thrown into different boxes, the first ball can be thrown into any one of the 20 boxes, the second ball can then be thrown into any one of the other 19 boxes, etc. Thus, there are $20 \cdot 19 \cdot 18 \cdots 9$ possible outcomes in the event. So the probability is $20!/8!20^{12}$.

8. There are 7^5 possible outcomes in the sample space. If the five passengers are to get off at different floors, the first passenger can get off at any one of the seven floors, the second passenger can then get off at any one of the other six floors, etc. Thus, the probability is

$$\frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3}{7^5} = \frac{360}{2401}.$$

9. There are $6!$ possible arrangements in which the six runners can finish the race. If the three runners from team A finish in the first three positions, there are $3!$ arrangements of these three runners among these three positions and there are also $3!$ arrangements of the three runners from team B among the last three positions. Therefore, there are $3! \times 3!$ arrangements in which the runners from team A finish in the first three positions and the runners from team B finish in the last three positions. Thus, the probability is $(3!3!)/6! = 1/20$.
10. We can imagine that the 100 balls are randomly ordered in a list, and then drawn in that order. Thus, the required probability in part (a), (b), or (c) of this exercise is simply the probability that the first, fiftieth, or last ball in the list is red. Each of these probabilities is the same $\frac{r}{100}$, because of the random order of the list.
11. In terms of factorials, $P_{n,k} = n!/[k!(n-k)!]$. Since we are assuming that n and $n-k$ are large, we can use Stirling's formula to approximate both of them. The approximation to $n!$ is $(2\pi)^{1/2}n^{n+1/2}e^{-n}$, and the approximation to $(n-k)!$ is $(2\pi)^{1/2}(n-k)^{n-k+1/2}e^{-n+k}$. The approximation to the ratio is the ratio of the approximations because the ratio of each approximation to its corresponding factorial converges to 1. That is,

$$\frac{n!}{k!(n-k)!} \approx \frac{(2\pi)^{1/2}n^{n+1/2}e^{-n}}{k!(2\pi)^{1/2}(n-k)^{n-k+1/2}e^{-n+k}} = \frac{e^{-k}n^k}{k!} \left(1 - \frac{k}{n}\right)^{-n-k-1/2}.$$

Further simplification is available if one assumes that k is small compared to n , that is $k/n \approx 0$. In this case, the last factor is approximately e^k , and the whole approximation simplifies to $n^k/k!$. This makes sense because, if $n/(n-k)$ is essentially 1, then the product of the k largest factors in $n!$ is essentially n^k .

1.8 Combinatorial Methods

Commentary

This section ends with an extended example called “The Tennis Tournament”. This is an application of combinatorics that uses a slightly subtle line of reasoning.

Solutions to Exercises

1. We have to assign 10 houses to one pollster, and the other pollster will get to canvas the other 10 houses. Hence, the number of assignments is the number of combinations of 20 items taken 10 at a time,

$$\binom{20}{10} = 184,756.$$

2. The ratio of $\binom{93}{30}$ to $\binom{93}{31}$ is $31/63 < 1$, so $\binom{93}{31}$ is larger.